

Data-Driven Discovery of the Dunkle Model: A Comparative Assessment

MECH 798M - Data-Driven Modeling in Science and Engineering

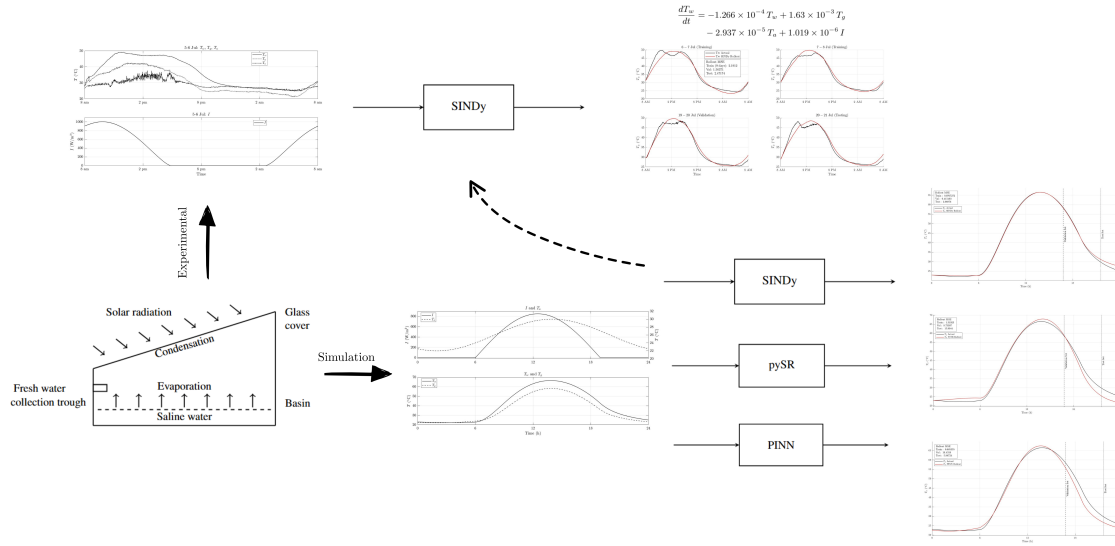
Project Report

Mansour Saliba

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Abstract

Solar stills are an innovative water treatment technology that is decentralized and cost effective yet suffers from limited productivity. This incentivized experimental and numerical research to enhance yield. However, computational research investigating the potential of physics-informed machine learning architectures is greatly lacking. This project aims to rediscover a simpler and more interpretable, yet still generalizable, Dunkle model adopting scientific machine learning architectures. The work is split into phases, one focusing on the heat transfer coefficients and another on the ODE describing the dynamics of water temperature. Results are promising in uncovering the underlying dynamics, particularly with experimental data, yet also reflect the limitations of the adopted architectures. This work provides an initial application of such numerical tools to model solar stills and future research must further investigate their potential with more complex systems of longer spans of data.



1. Introduction

While continuous supply of safe drinking water is a non-negotiable human right, at least 2 billion people globally lack access to essential water and sanitation services (WHO, 2025). This raises the need for innovative water treatment practices and desalination, through techniques such as Reverse Osmosis, is one effort in this aim, yet its cost hinders application. Solar desalination practices have gained popularity recently due to their minimal operational cost and applicability in resource-constrained settings (Ayoub & Malaeb, 2012). One such technology is the solar still (Figure 1), an insulated system containing brackish water treated through the processes of evaporation and

condensation, mirroring the hydrological cycle (Elango et al., 2015). Major research effort is continuously unfolding to improve the operation of solar stills, spanning both experimental and theoretical advances.

Experimentally, researchers have examined a plethora of parameters to improve solar still yield while maintaining cost-effectiveness. Such adjustments fall under four main categories: geometry, thermodynamics, hydrodynamics, and energy (Younis et al., 2022). Tested geometries include single slope, double slope, spherical, hemispherical, pyramidal, tubular, and conical (Sathyamurthy et al., 2016). Hydrodynamic enhancements primarily seek to enhance

evaporation through enlarging thin evaporative surface areas, and a main approach in this regard is that of a rotating drum (Ayoub et al., 2015). Moreover, improving the yield through adjusting the thermodynamics of the still is achieved by decreasing water depth, improving insulation, cooling the glass cover, and incorporating reflectors (Alwan et al., 2024). Lastly, energy-related enhancements seek to integrate energy storage material, either sensible or latent (Salunkhe & D., 2017). Yet, experimental research is often financially burdening and time consuming, increasing the need for accurate models simulating the thermal processes occurring within a solar still. A canonical advancement in this field is the model introduced by Dunkle back in 1961, derived through an empirical Nusselt-Rayleigh correlation of free convection of air in an enclosure (Dunkle, 1961; Elango et al., 2015).

Since then, major effort has been invested to overcome the limitations of Dunkle's model. Chen et al. (1984) proposed a model that operates over a wider range of Rayleigh number, Clark (1990) derived a model for a higher operating temperatures, Adhikari et al. (1990) suggested a simulation that is accurate for lower Grashof number, Kumar and Tiwari (1996) employed linear regression on experimental data to better estimate model coefficients, and Tsilingiris (2007) improved Dunkle's work by accounting for the properties of humid air (Elango et al., 2015). Significant work has also been implemented in employing computational fluid dynamic models to simulate solar stills (Rahbar & Esfahani, 2013). Despite such extensive research, Dunkle's model remains to be most adopted.

A major limitation in solar still research is the lack of integrating contemporary advancements in artificial intelligence, machine learning, and deep learning. Existing research is minimal and mostly applied through a data science lens that disregards thermodynamic principles at hand (Abdullah et al., 2024; Murugan et al., 2023). Such limitations point to a significant research gap between solar still modeling and the field of scientific machine learning.

Motivated by such extensive gaps, this project aims to rediscover Dunkle's governing equations through a comparative assessment of different physics-informed machine learning techniques.

2. Review

Dunkle's primary contribution is in estimating semi-empirical equations for convective heat flux h_c and evaporative heat flux h_e that enable estimation of freshwater yield m (Ayoub & Malaeb, 2012).

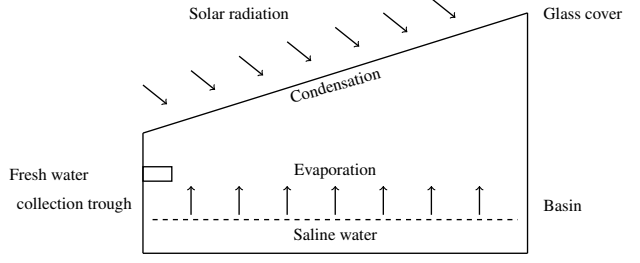


Figure 1. Single Slope Solar Still.

$$m = \frac{q_e}{h_{fg}} = \frac{h_e(T_w - T_g)}{h_{fg}} \quad (1)$$

where

$$h_e = 16.273 \times 10^{-3} \times h_c \left[\frac{(P_w - P_g)}{T_w - T_g} \right] \quad (2)$$

and

$$h_c = 0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g)(T_w + 273.15)}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \quad (3)$$

Such empirical equations are fundamental to the simulation of the thermal system. Through the first law of thermodynamics applied on the still's various components, one derives ordinary differential equations (ODEs) of the unfolding transient temperature profiles within the system. The parameter of interest in this project is temperature of the water T_w , which is expressed by an ODE that is characteristically simple (first order, quasi-linear) but mathematically complex.

$$\frac{dT_w}{dt} + aT_w = f(t) \quad (4)$$

The analytical solution exists and is derived from linear assumption over infinitesimal time steps.

$$T_w(t) = \frac{f}{a}(1 - e^{-at}) + T_w(t_0)e^{-at} \quad (5)$$

The terms a and f describe heat losses from the top and bottom of the still as well as the incoming energy from solar radiation. A full expansion of the ODE and its terms are outlined below, and an extensive treatment of the subject can be sought in Elango et al. and Tiwari & Sahota.

$$a = \frac{U_t + U_b}{m_w C_w}, \quad f = \frac{\alpha_{eff} I + (U_t + U_b)T_a}{m_w C_w} \quad (6)$$

The top heat loss coefficient U_t and bottom heat loss coefficient U_b are then expanded.

$$U_t = \frac{h_t U_g}{h_t + U_g}, \quad U_b = \frac{h_w h_b}{h_w + h_b} \quad (7)$$

Total heat fluxes include the previously outlined convective and evaporative heat coefficients in addition to the radiative heat coefficient outlined below,

$$h_t = h_c + h_e + h_r, \quad h_r = \epsilon_{eff} \sigma \frac{T_w^4 - T_g^4}{T_w - T_g} \quad (8)$$

Top and bottom losses also take into account the effect of material conduction.

$$U_g = \frac{\frac{K_g}{L_g} h_g}{\frac{K_g}{L_g} + h_g} \quad (9)$$

$$h_g = h_{gc} + h_{gr} = 2.8 + 3v + \epsilon_g \sigma \frac{T_g^4 - T_{sky}^4}{T_g - T_a} \quad (10)$$

$$h_b = \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7+3.8v}} \quad (11)$$

$$\alpha_{eff} = \alpha'_w + \alpha'_b \frac{h_w}{h_w + h_b} + \alpha'_g \frac{h_t}{h_t + U_g} \quad (12)$$

When plugged into the ODE, the full expanded term becomes:

$$\frac{dT_w}{dt} = -\frac{1}{m_w C_w} \left[\frac{\left[0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g)(T_w)}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \left(\frac{P_w - P_g}{T_w - T_g} \right) \right] + \epsilon_{eff} \times \sigma \times \left(\frac{T_w^4 - T_g^4}{T_w - T_g} \right) \right] \times \left[\frac{K_g}{L_g} \times \left[\frac{2.8+3v+\epsilon_g \times \sigma \times \left(\frac{T_g^4 - T_{sky}^4}{T_g - T_a} \right)}{2.8+3v+\epsilon_g \times \sigma \times \left(\frac{T_g^4 - T_{sky}^4}{T_g - T_a} \right)} \right] + \frac{h_w \times \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7+3.8v}}}}{\left[0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g)(T_w)}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \left(\frac{P_w - P_g}{T_w - T_g} \right) \right] + \epsilon_{eff} \times \sigma \times \left(\frac{T_w^4 - T_g^4}{T_w - T_g} \right) \right] + \left[\frac{K_g}{L_g} \times \left[\frac{2.8+3v+\epsilon_g \times \sigma \times \left(\frac{T_g^4 - T_{sky}^4}{T_g - T_a} \right)}{2.8+3v+\epsilon_g \times \sigma \times \left(\frac{T_g^4 - T_{sky}^4}{T_g - T_a} \right)} \right] + \frac{h_w + \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7+3.8v}}}} \right] (T_w - T_a)} \right] - \left[\alpha'_w + \alpha'_b \frac{h_w}{h_w + \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7+3.8v}}} + \alpha'_g \frac{0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g)(T_w)}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \left(\frac{P_w - P_g}{T_w - T_g} \right) \right] + \epsilon_{eff} \times \sigma \times \left(\frac{T_w^4 - T_g^4}{T_w - T_g} \right)}{\left[0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g)(T_w)}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \left(\frac{P_w - P_g}{T_w - T_g} \right) \right] + \epsilon_{eff} \times \sigma \times \left(\frac{T_w^4 - T_g^4}{T_w - T_g} \right) \right] + \left[\frac{K_g}{L_g} \times \left[\frac{2.8+3v+\epsilon_g \times \sigma \times \left(\frac{T_g^4 - T_{sky}^4}{T_g - T_a} \right)}{2.8+3v+\epsilon_g \times \sigma \times \left(\frac{T_g^4 - T_{sky}^4}{T_g - T_a} \right)} \right] + \frac{h_w + \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7+3.8v}}}} \right] } \right] I \right]$$

3. Methods

This project is implemented over two phases, starting with a discovery of h_c and h_e followed by the ODE of T_w in the second phase.

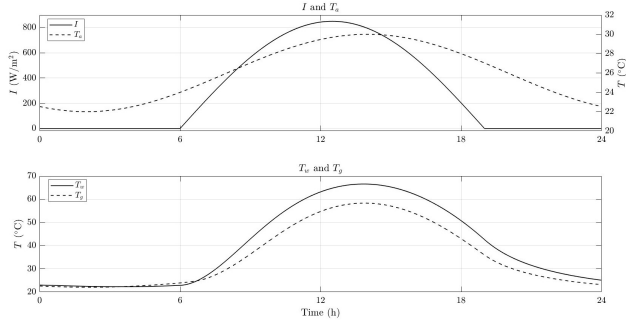


Figure 2. Simulated Data.

3.1. Dataset

Phase I relies on synthetic data while Phase II incorporates both synthetic and experimental data. To generate the synthetic data, the Dunkle model was forward solved using its analytical solution with a timestep $\Delta t = 1$ min and

initial conditions of $T_w(t_0) = 23^\circ\text{C}$ and $T_g(t_0) = 22.5^\circ\text{C}$, leading to 1,440 data points of T_w , T_g , h_c , h_e , and m . Climatic parameters, primarily I , v , and T_a , were simulated using simple equations mirroring Beirut's meteorological patterns. The GitHub repository included at the end of the report provides all scripts employed for data generation.

Empirical datasets collected at the American University of Beirut were also utilized in the project. They span 11 days of T_w , T_g , and T_a temperatures, with I data retrieved from Solcast. The data was adequately cleaned of erroneous measurements, and missing values were linearly interpolated and extrapolated. The size of the design matrix then amounts to $\mathbb{R}^{15,840 \times 4}$.

The synthetic data was split sequentially between 70% training, 20% validation, and 10% testing. Similarly, 9 days of experimental data were utilized for training while 1 day was employed for validation with the last day reserved for testing.

3.2. Phase I - Heat Coefficient Discovery

Three architectures will be compared, starting from a baseline neural network (NN) with no physics knowledge and then moving to symbolic regression (pySR) and sparse identification of non-linear dynamics (SINDy) (Brunton et al., 2016; Cranmer, 2023).

The NN is fully connected with 3 hidden layers, each of 64 neurons with tanh activations. The input to the network are vectors of T_w and T_g with the output being h_c and h_e . The Adam optimizer with a learning rate of 10^{-3} and a weight decay of 10^{-6} was employed. Epochs were capped at 3,000 with the possibility of early stopping based on increase of validation loss, and a mean squared error (MSE) loss was adopted.

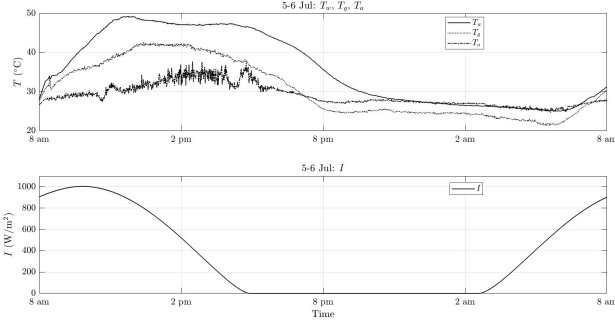


Figure 3. One Day of Experimental Data.

Two different approaches were implemented using pySR, the first tried uncovering h_c^3 and the second $\log(h_c)$. The search budget included 250 iterations, 16 populations, a population size of 40, and maximum expression size of 20. Possible operators include five (+, -, ×, ÷, ^) and the optimal equation is chosen based on minimum validation MSE in the h_c space. The difference in water and glass temperatures and pressures forms the feature set of both coefficients. In the case of h_e , h_c is also given as one of the features while in the case of h_c , P_w , T_w , and the compound term within the expression itself are provided as features.

SINDy was employed in three iterations. All trials employed sequentially thresholded least squares (STLSQ) with adaptive ridge stabilization (ridge baseline is 10^{-10}) and a threshold sweep of 20 values. The optimal sparse equation was selected by lowest MSE on the validation set with sparsity preference within 2% of the lowest error. Trial 1 constitutes the simplest approach whereby the feature library includes the compound term within h_c while the second and third trials try to uncover accurate models describing the coefficients from first and second order polynomial combinations, respectively, with the third trial also testing cubic roots of the feature library.

3.3. Phase II - ODE Discovery

This phase tests three architectures, SINDy, pySR, and physics-informed neural networks (PINNs) (Raissi et al., 2019), on the synthetic data. The goal is to investigate whether scientific machine learning techniques could uncover a simpler and more interpretable Dunkle model that avoids the convoluted expression described earlier, while remaining generalizable. The optimal performing model on the synthetic data is then utilized on the experimental dataset. All discovered ODEs were integrated forward and the solution was compared with the ground truth of T_w .

The feature library of the SINDy architecture comprised each of the variables I , T_a , T_w , and T_g with no bias term. The input comprised of the derivative of T_w computed numerically using Euler's scheme. The coefficients were estimated using the deterministic normal equations. Similar to phase I, STLSQ was adopted with adaptive ridge stabilization and a threshold sweep. Given this architecture, the purpose of SINDy is not so much as sparse regression but more so coefficient discovery mirroring multilinear regression.

The pySR architecture has the same target ODE and it uses the same variables, yet it searches over 80,000 candidate expressions rather than fitting coefficients. The possible operations are four (+, -, ×, ÷), and 10 populations with a size of 40, maximum expression of 18 and maximum depth of 8 were undertaken.

The PINN architecture differs from a regular NN in its incorporation within the loss function of a residual and initial condition losses. The common practice is to incorporate the expression of the differential equation within the residual, yet since the analytical solution is known in this case, it was inputted as the residual loss with five trainable coefficients to be discovered.

$$L = w_{\text{data}}L_{\text{data}} + w_{\text{res}}L_{\text{res}} + w_{\text{ic}}L_{\text{ic}} \quad (13)$$

$$L_{\text{data}} = \frac{1}{N} \sum_k \left(\hat{T}_{w,k} - T_{w,k} \right)^2 \quad (14)$$

$$L_{\text{res}} = \frac{1}{N-1} \sum_k \left(T_{w,k+1} - [c_1 F_k + (T_{w,k} - c_1 F_k) e^{-c_1 \Delta t_k}] \right)^2 \quad (15)$$

$$F_k = c_2 T_{a,k} + c_3 T_{g,k} + c_4 I_k + c_5 \quad (16)$$

$$L_{\text{ic}} = \left(\hat{T}_w(t_0) - 23 \right)^2 \quad (17)$$

4. Results

Figure 2 and Figure 3 showcase the simulated and experimental data of T_a , T_g , T_w , and I . The difference between the smoothness of the simulated data and the noise introduced by the K-Type Thermocouples can be clearly observed, providing a real world framework testing the modeling accuracy of the adopted architectures.

4.1. Phase I

Given the straightforward expression of h_e compared with the complex nature of h_c that includes a cubic root, it is expected for the former to be easily modeled compared with the latter. The NN baseline confirms this hypothesis as is clear in Figure 4. Moreover, as expected from a black box NN, it performs well on the training set yet deviates in testing, and the negative h_c values illustrate the lack of physics knowledge in the model.

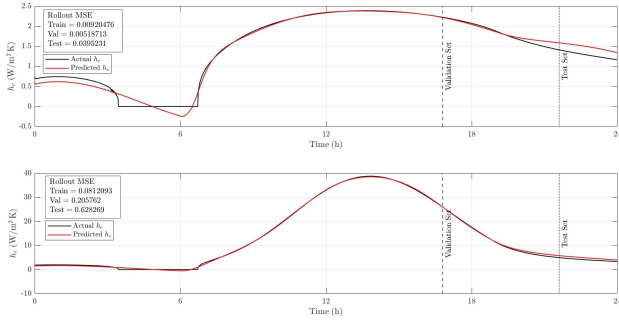


Figure 4. Actual vs. Predicted h_c and h_e - NN Baseline.

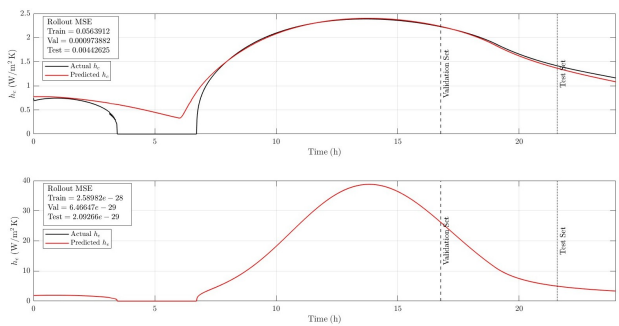


Figure 5. Actual vs. Predicted h_c and h_e - pySR.

Symbolic regression, in the case where h_c was cubed, showcases better test behavior yet worse training behavior whereby MSE for the latter increases from 0.009 to 0.056 and that of the former decreases from 0.039 to 0.004 (Figure 5). Such behavior reflects the primary contribution

of including the compound complicated term of h_c within the features of the architecture. Despite this, the architecture was still unable to recover the exact equation, as illustrated in the relatively poor training performance.

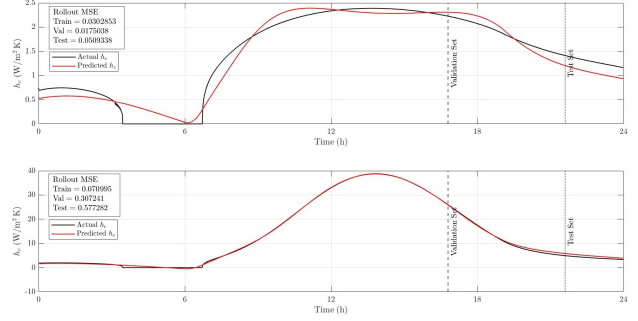


Figure 6. Actual vs. Predicted h_c and h_e - SINDy Trial 2.

When provided with the compound expression of h_c , the first trial of SINDy outperforms all other architectures as it is able to reinforce sparsity on extensive feature libraries and recover the exact equations. To investigate the algorithm further, first order polynomials of the variables in trial 2 showcase no sparsity (all terms included) and yet fail at modeling h_c well (Figure 6). Expanding the library to include second order polynomials, the model performs better, eliminating 59 expressions out of 90, maintaining positive h_c values yet showing slight deviations in validation and testing (Figure 7). The test MSE goes down from 0.05 to 0.004, reflecting considerable improvement due to the added expressions.

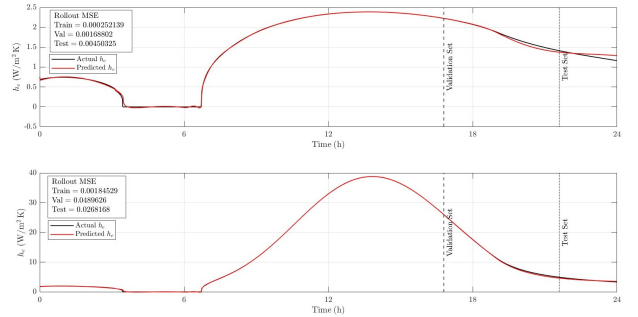


Figure 7. Actual vs. Predicted h_c and h_e - SINDy Trial 3.

4.2. Phase II

Prior to embarking on training the outlined architectures, this phase was initiated with an investigation of the solution's coefficients. As illustrated in Figure 8, the coefficients follow the temperature dynamic of a day's solar energy pro-

file in a parabolic shape with some deviations. Accordingly, the simplest model could be constructed by taking the average of the coefficients over time, resulting in coefficient values for T_w , T_a , and I . Integrated forward in time, the discovered ODE follows the general shape of the dynamics, yet with a significant MSE of almost 16 (Figure 9), indicating that the proposed machine learning architectures should be easily capable at recovering an accurate and simpler Dunkle model.

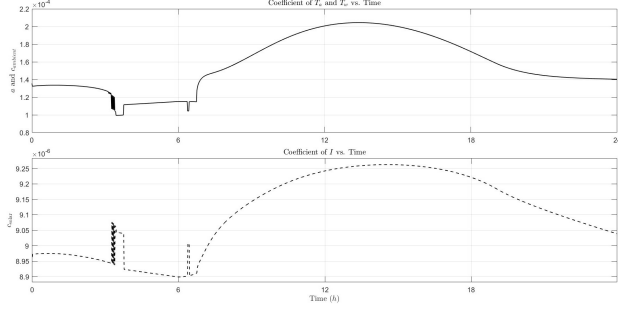


Figure 8. Change of the Actual Coefficients Over Time.

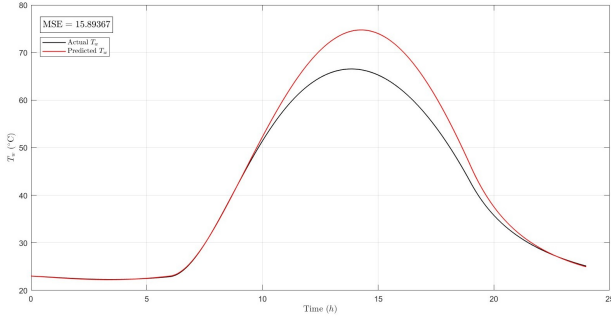


Figure 9. Actual vs. Predicted T_w - ODE of Average Coefficients.

This hypothesis is confirmed when implementing SINDy as the below discovered ODE recovers the shape of the dynamics and showcases relatively good generalizability, with a generalization gap of 3.89 (Figure 10).

$$\frac{dT_w}{dt} = -2.786 \times 10^{-5} T_w - 2.107 \times 10^{-4} T_g + 2.385 \times 10^{-4} T_a + 8.687 \times 10^{-6} I \quad (18)$$

Using pySR and PINN, the models still outperform the one derived from average coefficients yet they underperform compared with the one derived from SINDy, with generalization gaps of 14.58 and 5.2, respectively (Figures 11 and 12). Moreover, the equation derived from symbolic regression disregards the term T_w , an indicator of its poor performance.

$$\frac{dT_w}{dt} = 1.084 \times 10^{-5} (T_a \times T_g) + 0.00782 + 9.505 \times 10^{-6} I - 5.8834 \times 10^{-4} T_g \quad (19)$$

As for the PINN discovered coefficients, they reflect the different physical interactions of the temperature and solar radiation values with their negative or positive effects on T_w . It is worth noting here that the small coefficients of all models are a consequence of the energy source, the sun, which denotes that the thermal system is slowly rising and decaying.

$$T_{w,k+1} = 0.583 F_k + (T_{w,k} - 0.583 F_k) \times e^{-0.583 \Delta t_k} \quad (20)$$

where

$$F_k = -0.0369 T_{a,k} + 0.627 T_{g,k} + 0.00498 I_k - 0.1685 \quad (21)$$

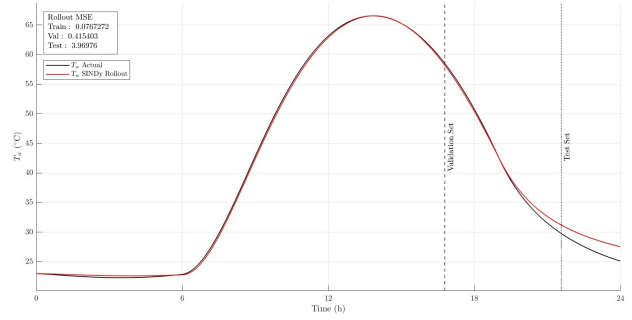


Figure 10. Actual vs. Predicted T_w - SINDy.

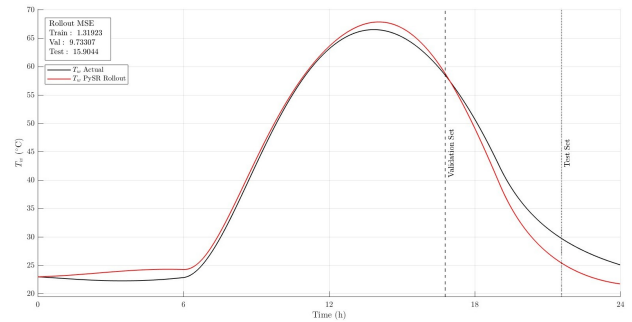


Figure 11. Actual vs. Predicted T_w - pySR.

Given that SINDy showcased the optimal performance during the synthetic phase, it was then adopted on the experimental data. All 9 days of training were stacked vertically in order to recover the below ODE that best describes the average dynamics across all days.

$$\frac{dT_w}{dt} = -1.266 \times 10^{-4} T_w + 1.63 \times 10^{-3} T_g - 2.937 \times 10^{-5} T_a + 1.019 \times 10^{-6} I \quad (22)$$

Surprisingly, experimental investigation showcases better results than all synthetic models, with a generalization gap of 0.28. Figure 13 showcases two days of training and the two days reserved for validation and testing. As observed, the model recovered the dynamic of rise and fall of the experimental data, despite its noisy nature, even with unseen data.

5. Discussion

This project presents the first treatment of physics-informed machine learning architectures applied to model the thermodynamic system of a conventional solar still. While the results showcase promising potential of the architectures in accurately recovering simplified dynamics overcoming convoluted first principle derivations, they also reflect some of the limitations of the adopted techniques.

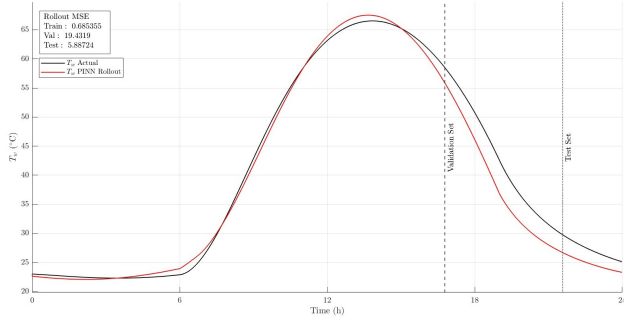


Figure 12. Actual vs. Predicted T_w - PINN.

Each phase presents disparate lessons that should be carefully adopted in future numerical studies. Primarily, phase I showcases the main limitation of such developed architectures in requiring a prerequisite knowledge of the physics. This limitation was outlined by Brunton et al. in their canonical SINDy paper and it is reflected in this project's results. For instance, discovery of h_c would not have been achieved had the empirical compound term discovered by Dunkle not been part of the feature library. Despite this, other numerically accurate yet physically deviating equations still showcased good generalizability. This illustrates the still limited ability of the field of scientific machine learning in uncovering novel accurate physics purely from data with no prior knowledge of the system. Furthermore, this aspect must be carefully approached when implementing such techniques as to

not fall in the trap of uncovering accurate models that disregard the underlying physical nature of the system at hand, as was the case with the third SINDy trial in Phase I.

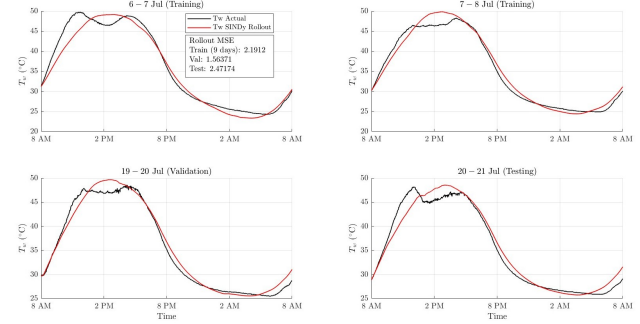


Figure 13. Actual vs. Predicted T_w - SINDy (Experimental).

On the other hand, phase II showcases the powerfulness and interpretability of the approaches, compared with black box architectures, when supplied with some physical knowledge. Indeed, the experimental results are greatly eye opening as to the yet uncovered potential of these numerical techniques within the field of solar stills. This is particularly so given the limited data employed, with just 9 days of training. Perhaps the main uncovered limitation is in their inability to capture the double peak phenomena around 2 pm. However, such models, when further developed and in systems much more complex than the conventional still, would greatly alleviate the experimental burden of solar still studies. Future research must investigate such architectures on other types of stills, optimize design parameters within the models' constraints, and invest effort in better understanding the physics of the discovered ML models. Proven accurate, the ultimate target of the solar still as a decentralized water treatment technology providing a safe supply of water to the most marginalized of communities would be greatly enhanced.

List of Abbreviations

- T_w : Basin water temperature ($^{\circ}\text{C}$)
- T_g : Inner glass temperature ($^{\circ}\text{C}$)
- T_a : Ambient temperature ($^{\circ}\text{C}$)
- T_{sky} : Sky temperature ($^{\circ}\text{C}$)
- P_w : Partial vapor pressure at T_w (N m^{-2})
- P_g : Partial vapor pressure at T_g (N m^{-2})
- m : Yield ($\text{kg m}^{-2} \text{s}^{-1}$)
- m_w : Mass of basin water (kg)
- C_w : Specific heat capacity of water ($\text{J kg}^{-1} \text{K}^{-1}$)
- I : Intensity of solar radiation (W m^{-2})
- v : Wind velocity (m s^{-1})
- σ : Stefan-Boltzmann constant ($\text{W m}^{-2} \text{K}^{-4}$)
- h_c : Convective heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
- h_e : Evaporative heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)

h_r : Radiative heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
 h_t : Total internal heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
 h_g : External glass heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
 h_{gc} : Convective heat transfer coefficient from glass to ambient ($\text{W m}^{-2} \text{K}^{-1}$)
 h_{gr} : Radiative heat transfer coefficient from glass to sky ($\text{W m}^{-2} \text{K}^{-1}$)
 h_w : Convective heat transfer coefficient from basin liner to water ($\text{W m}^{-2} \text{K}^{-1}$)
 h_b : Bottom conduction-convection heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
 h_{fg} : Latent heat of vaporization (J kg^{-1})
 U_t : Top loss coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
 U_b : Bottom loss coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
 U_g : Glass conduction loss coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
 K_g : Thermal conductivity of glass ($\text{W m}^{-1} \text{K}^{-1}$)
 L_g : Thickness of glass cover (m)
 K_{ins} : Thermal conductivity of insulation ($\text{W m}^{-1} \text{K}^{-1}$)
 L_{ins} : Thickness of insulation (m)
 q_e : Evaporative heat transfer rate (W m^{-2})
 α_{eff} : Effective solar absorptance (—)
 α'_w : Effective absorptance of basin water (—)
 α'_b : Effective absorptance of basin liner (—)
 α'_g : Effective absorptance of glass cover (—)
 ϵ_{eff} : Effective emissivity between water and glass (—)
 ϵ_g : Emissivity of glass cover (—)
 a : ODE decay coefficient (s^{-1})
 f : ODE forcing term ($^{\circ}\text{C s}^{-1}$)

GitHub Repository

https://github.com/MansourSaliba/Dunkle_ML

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